

Buying tips

Consider the following factors when buying a gigabit router:

- Performance
- Number of gigabit ports
- Multiple functionality
- Interoperability
- Built-in firewall and other security features
- Capability of the processor to handle the networking requirements
- Availability of ADSL, wireless, gigabit switching ports and gigabit WAN ports

a built-in intelligence called 'quality of service,' that helps them recognise applications that cannot tolerate even mild delays (such as streaming video) and prioritise the applications running in parallel such that the most sensitive applications experience the least latency. Moving to wireless-N devices significantly reduces disturbances during voice calls and distortions while streaming video and games.

It's already here

With so many advantages, manufacturers could not wait any longer to bring out the products on their anvil, and the market saw the launch of several 'pre-n' products based on 802.11n technology even before the Wi-Fi Alliance started certifying products based on draft 2.0 in 2007. Realising the market's eagerness and also confident about the maturity of the technology, the Alliance started testing and certifying products based on the draft 2.0 in 2007, even though the standard as such is expected to be ratified only in 2010.

Till date, over 500 products including several multimedia and enterprise products have been certified by the Alliance, and it is advised that consumers go only for certified products to ensure interoperability, security and backward compatibility when using multiple products from different manufacturers in the same network.

Wireless-N routers are doing well in the Indian market too. Suresh Balasubramanian, national sales director-India, Cisco Consumer Business

Group, explains that India is an extremely unique market in terms of its sensitivity to the cost of both Internet infrastructure as well as consumption. This means that consumers need to be able to have wireless connectivity with minimum investment and also need to manage their usage effectively. As a result, wireless-N routers are especially suited for the India market, both in terms of features as well as cost competitiveness.

Giga-speed with wireless-N

Wireless-N is not the only buzzword doing the rounds in the wireless router market. Without 'gigabit Ethernet' ports, a wireless-N router will be like a hawk without its talons!

A decade old standard, Gigabit Ethernet builds on the Ethernet protocol, scaling up its speed to 1000 megabits per second (Mbps) or 1 gigabit per second (Gbps), that is, 10/100 Mbps at the desktop level to 1000 Mbps in the data centre. Gigabit Ethernet is old, in fact, gigabit routers started arriving in the market as early as 1996. Now the term is used widely in the context of new wireless-N routers that come with gigabit Ethernet ports and enable these wireless routers to also connect to high-speed Ethernet connections.

Gigabit routers are in great demand in the Indian market, thanks to the customers' growing need for reliable high-speed data transmission. "Gigabit Ethernet is fast becoming the de-facto standard for metropolitan area networks (MANs), corporate wide area networks (WANs) and campus backbones as a cost-effective approach to traditional ATM and SONET rings.

Internet Protocol-based virtual private networks (VPNs) are also getting

Wireless-N devices are also capable of sending and receiving more than one communication signal at the same time, using multiple antennae

more popular, thereby increasing the demand for gigabit routers," says Subhashini Prabhakar, chief technology manager, Dax Networks. "This is true in developed as well as developing nations, since the utilisation of gigabit routers depends only on the application, the type of backbone, etc, and not constrained by slower Internet connections and other such factors."

Prem Nithin, senior technical consultant, Cisco, India & SAARC, explains that people use multi-function gigabit routers not just for connecting to the Internet from home but for telecommuting to work. Unified business-class wireless networking and remote working solutions are where gigabit routers are most useful, thanks to the inherent speed and reliability. "Teleworking is gaining great traction in India due to various factors," he says, hinting that it automatically increases the demand for wireless gigabit routers.

He also adds that the next stage in the evolution is 10 gigabit Ethernet, which is likely to become the norm in the near future.

Meeting consumer needs with new features

With great demand come great innovations. The network as such has great potential to improve the performance and functionality of almost every component of the IT environment—from servers to applications, middleware as well as end points. Technological developments and new features introduced in networking products, such as routers, are attempting to do just that.

"Routers have now matured to become intelligent network devices incorporating features such as content processing, VPN, firewall and load balancing. Wireless capabilities are also becoming popular, replacing the need for separate wireless access points for small office networks," says Balasubramanian.

Let us take a look at some of the interesting features in today's wireless-N gigabit routers.